

Juan Manuel Young Hoyos

SOFTWARE ENGINEER, FULL-STACK | TYPESCRIPT | SYSTEMS, REAL-TIME & WEBASSEMBLY

Medellín, Colombia · Open to relocating to Amsterdam (on-site) · English B2+

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Summary

Full-stack generalist who has shipped production systems in **TypeScript**, Go, Python and **C++** — including refactoring a **C++ WebAssembly** codebase — and who is happiest on problems that cut across the stack. I build **real-time, telemetry-driven** systems over **MQTT**, Kafka and message queues, design the architecture and the abstractions rather than just the features, and debug into unfamiliar layers to find the actual cause. Strong CS and **geometry/linear-algebra** foundation (spatial algorithms, 3D engines). I run **LLM agents in parallel** for building, review and investigation — and I have led delivery in chaotic environments against hard deadlines.

Skills

Languages	TypeScript , JavaScript, Python , Go , C++ , Nim · WebAssembly (refactored a legacy C++ WASM application)
Systems & Real-Time	Telemetry & real-time data , MQTT , Kafka, RabbitMQ, event-driven architecture , distributed & resumable queue systems, Linux (SSH, debugging), Docker
Frontend	React , React Native, Next.js · monitoring & control dashboards , state management, frontend performance (3.46 MB → 886 KB)
Geometry & Simulation	Geohash & hashmap spatial algorithms , 3D engines (Godot, Unity), physics/device simulation in Python, B.S. Computer Science & Engineering
AI Agents	Running multiple LLM agents in parallel (Claude Code) to build, review code, investigate incidents and generate tests — validating output with engineering judgement
Familiar	Rust, Kubernetes, WebSockets & low-level networking (TCP/IP, UDP)

Experience

Okorum Technologies

Medellín, Colombia (Hybrid)

TECHNICAL LEAD

Dec. 2025 – Present

- Chose the core architecture and abstractions for a national-scale platform and shipped it **ahead of a hard government deadline** — a distributed **TypeScript** monorepo whose stages coordinate over **message queues**, with the **React** front end in the same repo.
- Made an unreliable physical-world dependency survivable by engineering **fault tolerance and resumability** into the queue system, so in-flight work recovers when the upstream is slow or down — plus **Redis** caching and **telemetry** (logs, traces, metrics via Grafana/Loki) to see what every stage is doing.
- Drove problems start-to-finish with a cross-functional team, without external project management.

ArcDesign

Spain (Remote)

FULL STACK DEVELOPER (PART-TIME)

Jul. 2023 – Mar. 2025

- Designed a spatial algorithm using **hashmaps and geohashes** that improved performance and scalability, and drove the architecture around **MQTT**, running device **simulations in Python** before integrating the logic into a client for **near real-time** behaviour.

Grisú

Colombia (Remote)

FULL STACK DEVELOPER & GAME DEVELOPER (SELF-EMPLOYED)

Jul. 2023 – Present

- Built and launched **React Native** apps and a hyper-casual game in **Godot** — 3D/2D engine work involving transforms and geometry — and compressed a 4-month plan into weeks on **EcoDrive**, an event-driven system in **Kafka** and Python.

Design Systems Inno

Medellín, Colombia (Hybrid)

PROBLEM SOLVER — PLATFORM & AUTOMATION

Dec. 2022 – Jun. 2023

- Architected a decoupled, asynchronous system on **RabbitMQ** where clients track long-running jobs by ticket ID instead of blocking, with internal APIs in **Go** and **Python** on MongoDB.
- Cut React/Angular bundle sizes from 3.46 MB to **886 KB**, improved Docker images, and paired with teammates to fix critical bugs before delivery.

Aligo · Geta Club Play

Colombia

EARLIER EXPERIENCE — SYSTEMS, SECURITY & QA

Feb. 2021 – Oct. 2022

- Refactored and retrofitted a legacy **C++ WebAssembly** application, improving code quality by **45%**; wrote low-level tooling in **Go**, **Python** and Nim; reverse-engineered Android binaries; reduced known vulnerabilities **50%** in 4 months.

Education

EAFIT University

Medellín, Colombia

B.S. IN COMPUTER SCIENCE AND ENGINEERING

Jan. 2018 – Dec. 2025

- Full scholarship “Ser Pilo Paga” (national merit-based award).